

Last week: One-loop planar operator mixing in the $SU(2)$ sector of $N=4$ SYM

$$\Delta = L + \frac{\lambda}{8\pi^2} \sum_{k=1}^L (Z_{k,k+1} - P_{k,k+1})$$

Hamiltonian of $XXZ_{1/2}$

$$H_{XXZ} \sim \frac{d}{d\lambda} \log \text{tr} \left(L_{L_2}(\lambda) \cdots L_{L_1}(\lambda) \right) \Big|_{\lambda = \frac{i}{2}}$$

$= T(\lambda)$, transfer matrix

This week : How to find (all) eigenvalues of H_{XXX}
by solving Bethe Ansatz Equations

Singular solutions of BAE

Separation of Variables

cf. XXX spin chain is related to many other models :

$\left\{ \begin{array}{l} \text{6-vertex, sine-Gordon, massive Thirring} \\ c=1 \text{ CFT at self-dual point, ...} \end{array} \right\}$

§ Transfer matrix eigenvalues

[Faddeev, 9605187]

$$L_{n,\alpha}(\lambda) = \begin{pmatrix} \lambda + iS_n^3 & iS_n^- \\ iS_n^+ & \lambda - iS_n^3 \end{pmatrix}$$

$$S^+|\uparrow\rangle = 0 \Rightarrow \text{Define Bethe vacuum } |0\rangle = |\uparrow\uparrow \dots \uparrow\rangle$$

$$\text{In terms of } N=4 \text{ SYM: } \{\uparrow, \downarrow\} \longleftrightarrow \{z, w\}$$

$$|0\rangle \longleftrightarrow \pi z^L$$

$$L_{n,\alpha}(\lambda) |0\rangle = \begin{pmatrix} \lambda + iS_n^3 & * \\ 0 & \lambda - iS_n^3 \end{pmatrix}$$

Monodromy matrix :

$$M_a(\lambda)_{ij} = \left(\mathcal{L}_{L,a}(\lambda) \cdots \mathcal{L}_{1,a}(\lambda) \right)_{ij} \equiv \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix}$$

$i, j = 1, 2$, A, B, C, D : $2^L \times 2^L$ matrices

$$M_a(\lambda)_{ij} |0\rangle = \begin{pmatrix} \lambda + \frac{i}{2} & * \\ 0 & \lambda - \frac{i}{2} \end{pmatrix}^L |0\rangle$$

$$\Rightarrow A(\lambda) |0\rangle = \left(\lambda + \frac{i}{2} \right)^L |0\rangle$$

$$D(\lambda) |0\rangle = \left(\lambda - \frac{i}{2} \right)^L |0\rangle$$

$$\frac{d}{d\lambda} \log (A(\lambda) + D(\lambda)) \Big|_{\lambda = \frac{i}{2}} = (\text{const}) \quad \text{vacuum energy}$$

For excited states, consider RTT relations

(In my notation, $RMM = MMR$)

$$R_{a_1 a_2}(\lambda - \mu) M_{a_1}(\lambda) M_{a_2}(\mu) = M_{a_2}(\mu) M_{a_1}(\lambda) R_{a_1 a_2}(\lambda - \mu)$$

$$R_{ab}(\lambda - \mu) = (\lambda - \mu) I_{ab} + i P_{ab}$$

$$= \begin{pmatrix} \lambda - \mu + i & 0 & 0 & 0 \\ 0 & \lambda - \mu & i & 0 \\ 0 & i & \lambda - \mu & 0 \\ 0 & 0 & 0 & \lambda - \mu + i \end{pmatrix} \begin{matrix} \uparrow\uparrow \\ \uparrow\downarrow \\ \downarrow\uparrow \\ \downarrow\downarrow \end{matrix}$$

$\{\uparrow\uparrow, \uparrow\downarrow, \downarrow\uparrow, \downarrow\downarrow\}$ is natural basis of $V_{a_1} \otimes V_{a_2}$

$$M_a(\lambda)_{ij} = i \xrightarrow[\substack{V_L \otimes V_{L-1} \cdots \otimes V_1}]{\uparrow \uparrow \cdots \uparrow} j = \begin{pmatrix} A(\lambda) & B(\lambda) \\ C(\lambda) & D(\lambda) \end{pmatrix}$$

$$M_{a_1}(\lambda) \otimes M_{a_2}(\mu) = \begin{pmatrix} A(\lambda)A(\mu) & A(\lambda)B(\mu) & B(\lambda)A(\mu) & B(\lambda)B(\mu) \\ A(\lambda)C(\mu) & A(\lambda)D(\mu) & \cdot & \cdot \\ C(\lambda)A(\mu) & \cdot & \cdot & D(\lambda)B(\mu) \\ C(\lambda)C(\mu) & \cdot & \cdot & D(\lambda)D(\mu) \end{pmatrix}$$

$$M_{a_2}(\mu) \otimes M_{a_1}(\lambda) = \begin{pmatrix} A(\mu) A(\lambda) & B(\mu) A(\lambda) & A(\mu) B(\lambda) & B(\mu) B(\lambda) \\ C(\mu) A(\lambda) & D(\mu) A(\lambda) & C(\mu) B(\lambda) & D(\mu) B(\lambda) \\ A(\mu) C(\lambda) & \cdot & \cdot & B(\mu) D(\lambda) \\ A(\mu) C(\lambda) & \cdot & \cdot & D(\mu) D(\lambda) \end{pmatrix}$$

note that tensor structure is different, because
we need to keep track of (a_1, a_2) indices

$RMM = MMR \Rightarrow 16$ equations for $2^L \otimes 2^L$ matrices

$$\begin{pmatrix} \lambda - \mu + i & \\ & \ddots \end{pmatrix} \begin{pmatrix} A(\lambda) A(\mu) & \\ & \ddots \end{pmatrix} = \begin{pmatrix} A(\mu) A(\lambda) & \\ & \ddots \end{pmatrix} \begin{pmatrix} \lambda - \mu + i & \\ & \ddots \end{pmatrix}$$

$$(1,1) \text{ component} \Rightarrow A(\lambda) A(\mu) = A(\mu) A(\lambda)$$

$$\text{similarly } [B(\lambda), B(\mu)] = [C(\lambda), C(\mu)] = [D(\lambda), D(\mu)] = 0$$

$$\left\{ \begin{array}{ll} A + D \rightarrow & \text{eigenvalues of } T(\lambda) \\ B \rightarrow & \text{creation op.} \\ C \rightarrow & \text{annihilation op.} \end{array} \right.$$

$$\text{We need } XB = * BX + * BX \quad \text{for } X = A, D$$

Non-trivial components : $(1,3)$ and $(2,4)$

$$\begin{cases} (\lambda - \mu + i) B(\lambda) A(\mu) = : B(\mu) A(\lambda) + (\lambda - \mu) \underline{A(\mu) B(\lambda)} \\ (\lambda - \mu) \underline{D(\lambda) B(\mu)} + : B(\lambda) D(\mu) = (\lambda - \mu + i) B(\mu) D(\lambda) \end{cases}$$

Ansatz for excited states (Bethe states) :

$$|\vec{u}\rangle = B(u_1) B(u_2) \dots B(u_n) |0\rangle$$

Evaluate transfer matrix using commutation relations :

$$(A(v) + D(v)) B(u_1) \dots B(u_n) |0\rangle$$

$$A(v) B(u_1) = \frac{v - u_1 - i}{v - u_1} B(u_1) A(v) + \frac{i}{v - u_1} B(v) A(u_1)$$

$$A(v) B(u_2) = \dots$$

$$A(u_1) B(u_2) = \dots$$

Commutation relations generate 2^M terms, summarized as

$$A(v) |\vec{u}\rangle = \left(\prod_{j=1}^M \frac{v - u_j - i}{v - u_j} \right) B(u_1) \dots B(u_M) \cdot A(v) |0\rangle$$

$$= \left(v + \frac{i}{2} \right)^L |0\rangle$$

$$+ \sum_{k=1}^M M_k(v) \cdot B(v) B(u_1) \dots (\text{no } B(u_k)) \dots B(u_M) A(u_k) |0\rangle$$

$$\text{unwanted term, not } |\vec{u}\rangle = \left(u_k + \frac{i}{2} \right)^L |0\rangle$$

The coefficient M_k :

The $k=1$ term comes from

$$\begin{cases} A(v) B(u_1) = f(v-u_1) B(u_1) A(v) + \underline{g(v-u_1) B(v) A(u_1)} \\ A(u_1) B(u_j) = \underline{f(u_1-u_j) B(u_j) A(u_1)} + g(u_1-u_j) B(u_1) A(u_j) \end{cases} \text{ for } j \geq 2$$

$$\Rightarrow M_1(v) = \frac{i}{v-u_1} \prod_{j=2}^M \frac{u_1 - u_j - i}{u_1 - u_j}$$

$k \geq 2$ are complicated, but $B(u_1) \cdots B(u_M) |0\rangle$ is

invariant under permutation $u_j \rightarrow u_{\sigma(j)}$, $\sigma \in S_M$

Commutation relations for DB;

$$D(v) B(u_1) = \frac{v - u_1 + i}{v - u_1} B(v) D(u_1) - \frac{i}{v - u_1} B(u_1) D(v)$$

$$D(v) |\vec{u}\rangle = \left(\prod_{j=1}^M \frac{v - u_j + i}{v - u_j} \right) \underbrace{B(u_1) \cdots B(u_M)}_{= (v - \frac{i}{2})^L |0\rangle} D(v) |0\rangle$$

$$+ \sum_k \tilde{M}_k(v) \underbrace{B(v) B(u_1) \cdots (no\ B(u_k)) \cdots B(u_M)}_{\text{unwanted term}} \underbrace{D(u_k)}_{(u_k - \frac{i}{2})^L |0\rangle} |0\rangle$$

Therefore

$$\begin{aligned}
 & (A(v) + D(v)) |\vec{u}\rangle \\
 &= \left\{ \left(v + \frac{i}{2}\right)^L \prod_{j=1}^M \left(\frac{v - u_j - i}{v - u_j}\right) + \left(v - \frac{i}{2}\right)^L \prod_{j=1}^M \left(\frac{v - u_j + i}{v - u_j}\right) \right\} |\vec{u}\rangle \\
 &+ \sum_{k=1}^M \left\{ M_k(v) \left(u_k + \frac{i}{2}\right)^L + \tilde{M}_k(v) \left(u_k - \frac{i}{2}\right)^L \right\} \\
 &B(v) B(u_1) \dots (no B(u_k)) \dots B(u_M) |0\rangle
 \end{aligned}$$

$$\tilde{M}_1(v) = \frac{-i}{v - u_1} \prod_{j=2}^M \frac{u_1 - u_j + i}{u_1 - u_j} \propto \frac{1}{v - u_1}$$

Can derive Bethe Ansatz Equations (BAE) in two ways

1) Unwanted term = 0.

$$\left(u_1 + \frac{i}{2}\right)^L \prod_{j=2}^M \frac{u_1 - u_j - i}{u_1 - u_j} - \left(u_1 - \frac{i}{2}\right)^L \prod_{j=2}^M \frac{u_1 - u_j + i}{u_1 - u_j} = 0$$

$$\Rightarrow \left(\frac{u_k + \frac{i}{2}}{u_k - \frac{i}{2}} \right)^L = \prod_{j \neq k}^M \frac{u_k - u_j + i}{u_k - u_j - i}$$

for $k=1, 2, \dots, M$

2) Transfer matrix $T(v) = A(v) + D(v)$

must be a polynomial of v in degree L

$$T(v) = \left(v + \frac{i}{2}\right)^L \prod_{j=1}^M \frac{v - u_j - i}{v - u_j} + \left(v - \frac{i}{2}\right)^L \prod_{j=1}^M \frac{v - u_j + i}{v - u_j}$$

potentially singular at $v = u_k$; $\text{Res } T(v) = 0$

$$0 = -i \left(u_k + \frac{i}{2}\right)^L \prod_{j \neq k} \frac{u_k - u_j - i}{u_k - u_j} + i \left(u_k - \frac{i}{2}\right)^L \prod_{j \neq k} \frac{u_k - u_j + i}{u_k - u_j}$$

$\Rightarrow$ Same equation as before

Conserved charges:

$$1) \quad U \sim T(\lambda = \frac{i}{2}) \sim \prod_{j=1}^M \left(\frac{u_j + i/2}{u_j - i/2} \right) \equiv \prod_{j=1}^M e^{i p_j}$$

shift operator $U \longleftrightarrow$ total momentum P

$$2) \quad E \sim \left. \frac{d}{d\lambda} \log T(\lambda) \right|_{\lambda = \frac{i}{2}} \sim \sum_{j=1}^M \frac{1}{(u_j + \frac{i}{2})(u_j - \frac{i}{2})}$$

$$3) \quad \text{Higher charges: } Q_r \equiv \sum_j \left\{ \frac{1}{(u_j + i/2)^{r-1}} - \frac{1}{(u_j - i/2)^{r-1}} \right\}$$

$SU(2)$ symmetry :

Taking $\lambda \rightarrow \infty$ in RTT relations,

Using $T(\lambda) = \begin{pmatrix} \lambda^N + i\lambda^{N-1}S^3 + \dots & i\lambda^{N-1}S^- + \dots \\ i\lambda^{N-1}S^+ + \dots & \lambda^N + i\lambda^{N-1}S^3 + \dots \end{pmatrix}$

$$A(\lambda)B(\mu) = \underbrace{\frac{\lambda - \mu - i}{\lambda - \mu} B(\mu)A(\lambda)}_{= 1 - \frac{i}{\lambda} + \dots} + \underbrace{\frac{i}{\lambda - \mu} B(\lambda)A(\mu)}_{\mathcal{O}(\lambda^{N-2})}$$

Compare $\mathcal{O}(\lambda^{N-1})$:

$$iS^3 B(\mu) = iB(\mu)S^3 - iB(\mu) \Rightarrow [S^3, B(\mu)] = -B(\mu)$$

Similarly, (2,3) component of RTT relation:

$$(u-v) B(u) C(v) + i D(u) A(v) = i D(v) A(u) + (u-v) C(v) B(u)$$

large u limit & collect $O(u^L)$ using $C \rightarrow i u^{L-1} S^+$

$$[S^+, B(u)] = A(u) - D(u)$$

For the Bethe vacuum, $|0\rangle = |\uparrow\uparrow\cdots\uparrow\rangle$

$$S^3 |0\rangle = \sum_{n=1}^L S_n^3 |0\rangle = \frac{L}{2} |0\rangle$$

$$S^3 |\vec{u}\rangle = \left(\frac{L}{2} - M\right) |\vec{u}\rangle$$

Thus (regular) Bethe states are the $SU(2)$ HWS

$$S^+ |\vec{u}\rangle = 0.$$

Add $u_{M+1} = \infty$ to a solution of BAE $\{u_k\}$

$$\left(\frac{u_k + i/2}{u_k - i/2} \right)^L = \prod_{j \neq k}^M \frac{u_k - u_j + i}{u_k - u_j - i} \frac{u_k - \infty + i}{u_k - \infty - i} = 1$$

$|u'\rangle \equiv \underline{B(\infty)} B(u_1) \cdots B(u_M) |0\rangle$ is still a solution

$= :v^{L-1} S^- (v \rightarrow \infty) ; \quad SU(2) \text{ descendants}$

$$S^+ |\vec{u}\rangle = \sum_{k=1}^M B(u_1) \dots B(u_{k-1}) \left(A(u_k) - D(u_k) \right) \dots B(u_M) |0\rangle$$

bring right

$$\equiv \sum_{l=1}^M C_l B(u_1) \dots (\text{no } B(u_l)) \dots B(u_M) |0\rangle$$

Using RTT relations,

$$(A(u_1) - D(u_1)) B(u_2) = B(u_2) \left(\frac{u_1 - u_2 + i}{u_1 - u_2} A(u_1) - \frac{u_1 - u_2 - i}{u_1 - u_2} D(u_1) \right) + \dots$$

$$C_l = \prod_{j=2}^M \frac{u_1 - u_j + i}{u_1 - u_j} \left(u_1 + \frac{i}{2} \right)^L - \prod_{j=2}^M \frac{u_1 - u_j - i}{u_1 - u_j} \left(u_1 - \frac{i}{2} \right)^L = 0$$

permutation symmetry $\Rightarrow C_l = 0$ from BAE for u_l

§ Singular solutions :

[Marboe, Volin, 1608.06504]

Bethe states $|\vec{u}\rangle \Rightarrow$ BAE for $\{u_k\}$

($\sim$ PBC for particles on length- L circle)

$\Leftarrow$ is not true!

e.g.

$$\left(\frac{u_k + i/2}{u_k - i/2} \right)^L = \prod_{j \neq k} \frac{u_k - u_j + i}{u_k - u_j - i}$$

if $(u_1, u_2) \approx (i/2, -i/2)$, we get $\infty = \infty$

can regularize by ϵ , but needs $\mathcal{O}(\epsilon^L)$ term

There are other meaningless solutions like $u_j = u_k$

'meaningless' because Gaudin norm vanishes

$$\langle u | u \rangle = \det_{j,k} \left(\frac{\partial}{\partial u_j} \log BAE_k \right)$$

But there can be "almost degenerate" solutions
for twisted boundary conditions

$$e^{i\phi} \left(\frac{u_k + i/2}{u_k - i/2} \right)^L = e^{i\phi + iLp_k} = \prod_{j \neq k} \frac{u_k - u_j + i}{u_k - u_j - i}$$

Numerically, better to solve Baxter TQ relations

$$T(u) Q(u) = \phi(u - \frac{i}{2}) Q(u+i) + \phi(u + \frac{i}{2}) Q(u-i)$$

$$\phi(u) = u^L, \quad Q(u) = \prod_{j=1}^M (u - u_j)$$

polynomiality of $T(u) \Rightarrow$ BAE

But Baxter TQ relations with $(u_1, u_2) = (\frac{i}{2}, -\frac{i}{2})$

may contain "unphysical" solutions (overcounting)

$\leadsto$ need to solve QQ-system for a complete sol.

Remark : in coordinate Bethe Ansatz,

$$\psi(p_1, p_2) = \sum_{n_1 < n_2} \left\{ e^{ip_1 n_1 + ip_2 n_2} + S(p_2, p_1) e^{ip_1 n_2 + ip_2 n_1} \right\}$$

$$\text{tr} \left(Z \cdots \hat{n}_1 \cdots W \cdots \hat{n}_2 \cdots W \cdots Z \right)$$

From trace cyclicity, ψ is invariant under the shift

$$(n_1, n_2, \dots) \mapsto (n_1+1, n_2+1, \dots) \Rightarrow e^{iP_{\text{tot}}} = 1$$

"momentum quantization" $e^{iL P_{\text{tot}}} = 1$

$\leadsto$ take a product of all BAE, $\prod_k e^{iL p_k} = 1$

§ Separation of Variables

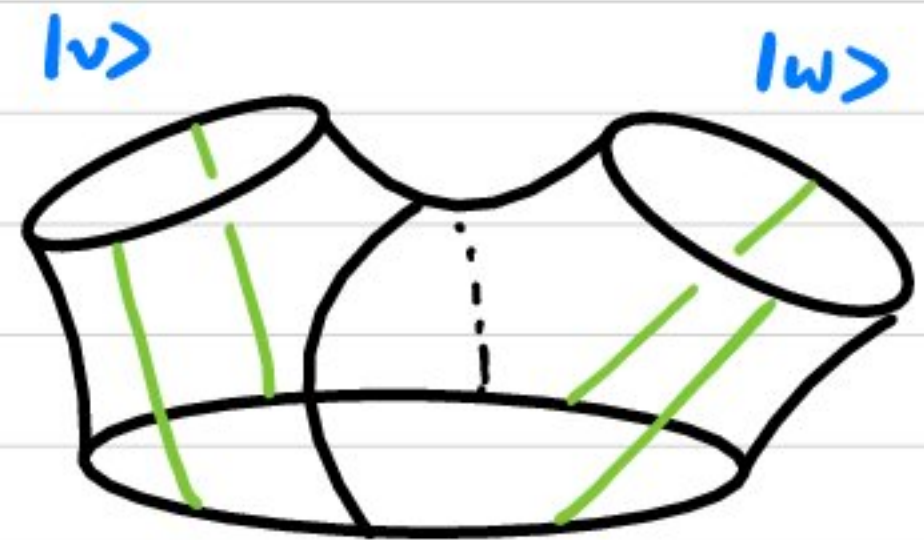
3pt functions of $N=4$ SYM:

- Split Bethe state $|u\rangle = |u'\rangle \otimes |u''\rangle$
- take inner product $\langle \mathcal{O}_u \mathcal{O}_v \mathcal{O}_w \rangle \sim \sum_{u=u' \cup u''} \langle v|u'\rangle \langle w|u''\rangle$

$$L = L_1 + L_2, \quad M = M_1 + M_2$$

$$\{u_j\}_{j=1}^M = \{u_j\}_{j=1}^{M_1} \cup \{u_j''\}_{j=M_1+1}^{M_1+M_2}$$

Solves BAE for L , not L_1
 $\Rightarrow$ off-shell



$$|u\rangle = |u'\rangle \otimes |u''\rangle$$

For (on-shell) Bethe states

$$\langle u|v \rangle \propto \delta(\{u\}, \{v\}) \times (\text{Gaudin det})$$

For off-shell Bethe states

$$\langle u|v \rangle \sim (\text{Slavnov determinant})$$

← Separation of Variables (SoV) basis

$$\langle x|u \rangle = \prod_{j=1}^L \prod_{k=1}^M (x_j - u_k)$$

What is this X ?

cf. [Ryan Volin 1810.10996]

[Gromov Levkovich-Maslyuk, Sizov, 1610.08032]

SoV basis :

complete basis which diagonalizes (twisted) B-operator

$$\langle x | B(u) = \prod_{j=1}^L (u - x_j)$$

$B(u)$ is nilpotent (not diagonalizable) in periodic chain

Twist : $T_a^K(u) = \text{tr} (K \mathcal{L}_{L,a}(u) \cdots \mathcal{L}_{1,a}(u))$

$$= \begin{pmatrix} A^K(u) & B^K(u) \\ C^K(u) & D^K(u) \end{pmatrix}$$

$$K \sim \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}$$

Expand $B^k(u) = \sum_{n=0}^L u^n B_n^k$

$$B_n^k \sim \text{diag} \{ b_n^{(1)}, b_n^{(2)}, \dots, b_n^{(2^L)} \}$$

For each eigenvalue $b^{(\alpha)}(u) = \sum_{n=0}^L u^n b_n^{(\alpha)} \quad (\alpha=1, \dots, 2^L)$

define the polynomial roots $b^{(\alpha)}(u) = \prod_{j=1}^L (u - x_j^{(\alpha)})$

Left eigenvector $\langle x^{(\alpha)} | B^k(u) = \langle x^{(\alpha)} | \prod_j (u - x_j^{(\alpha)})$

Off-shell Bethe state : $B(u_1) \cdots B(u_n) |0\rangle \equiv |u\rangle$

Wave-function for one magnon : $\psi(x|u) \equiv \prod_{j=1}^L (u - x_j)$

Some people introduces "quantum position operator"

$$\langle x_1, x_2, \dots, x_L | \hat{x}_j = \langle x_1, x_2, \dots, x_L | x_j$$

but $\{\hat{x}_j\}$ is complicated ($\hat{x}_j \sim (E_{11})_j$ for simple case)

Spectrum of $\hat{x}_j$ is $x_j = \theta_j \pm \frac{i}{2}$ (θ_j : inhomogeneity)

Sol points

Evaluate transfer matrix:

$$\langle x^{(\alpha)} | T(v) | u \rangle = \langle x | (A(v) + D(v)) B(u_1) \cdots B(u_M) | 0 \rangle$$

RTT relations:

$$(A(v) + D(v)) B(u_1) = \frac{B(u_1)}{v - u_1} \left\{ (v - u_1 - i) A(v) + (v - u_1 + i) D(v) \right\} \\ + \frac{B(v)}{v - u_1} \left\{ i A(u_1) - i D(u_1) \right\}$$

unwanted term vanishes if $v = x_j^{(\alpha)}$

$$\langle x^{(\alpha)} | B(v) = \langle x^{(\alpha)} | \prod_{j=1}^L (v - x_j^{(\alpha)})$$

$$\langle x^{(\alpha)} | T(x_j^{(\alpha)}) | u \rangle = \psi(x^{(\alpha)} | u_1) \cdots \psi(x^{(\alpha)} | u_M)$$

$$\left\{ \frac{Q(v-i)}{Q(v)} \left(v + \frac{i}{2}\right)^L + \frac{Q(v+i)}{Q(v)} \left(v - \frac{i}{2}\right)^L \right\}_{v=x_j^{(\alpha)}}$$

Using

$$\Psi(x) \equiv \prod_{k=1}^M \psi(x | u_k) = \prod_k^M \prod_{j=1}^L (u_k - x_j) = (-1)^{LM} \prod_{j=1}^L Q(x_j)$$

$$\begin{aligned} \Psi(x) \frac{Q(x_j^{(\alpha)} \mp i)}{Q(x_j^{(\alpha)})} &= (-1)^{LM} \prod_{l \neq j} Q(x_l^{(\alpha)}) \cdot Q(x_j \mp i) \\ &= \Psi(x_1, \dots, x_j \mp i, \dots, x_L) \end{aligned}$$

Therefore

$$\begin{aligned} \langle x^{(a)} | T(x_j^{(a)}) | u \rangle &= (x_j^{(a)} + \frac{i}{2})^L \Psi(x_1 \cdots, x_{j-i}, \cdots x_L) \\ &+ (x_j^{(a)} - \frac{i}{2})^L \Psi(x_1 \cdots, x_{j+i}, \cdots x_L) \quad (j=1, \cdots, L) \end{aligned}$$

For an on-shell state :

$$\begin{aligned} \langle x^{(a)} | T | u \rangle &= (A(x_j^{(a)}) + D(x_j^{(a)})) \Psi(x^{(a)}) \\ &\equiv t(x_j^{(a)}) \Psi(x^{(a)}) \end{aligned}$$

$$\Psi(x_1 \cdots x_L) = (-1)^{LM} \prod_j Q(x_j) \rightarrow \text{equation factorizes for each } j$$

Finding Sol points (Spectrum of B-operator)

(see also ; Kazama - Komatsu - Nishimura 1304, 5011)

If you introduce inhomogeneity

$$T(v) = \text{tr} \left(\mathcal{L}_{L,a}(v - \theta_L) \cdots \mathcal{L}_{1,a}(v - \theta_1) \right)$$

transfer matrix eigenvalue with θ 's :

$$\left\{ \begin{array}{l} t(v) = \phi(v + \frac{i}{2}) \frac{Q(v-i)}{Q(v)} + \phi(v - \frac{i}{2}) \frac{Q(v+i)}{Q(v)} \\ \phi(v) = \prod_{j=1}^L (v - \theta_j) \end{array} \right.$$

Difference equation for the total wave function :

$$t(x_j^{(\alpha)}) \bar{\Psi}(x_1^{(\alpha)}, \dots, x_L^{(\alpha)})$$

$$= \phi(x_j^{(\alpha)} + \frac{i}{2}) \bar{\Psi}(x_1^{(\alpha)}, \dots, x_j^{(\alpha)} - i, \dots, x_L^{(\alpha)})$$

$$+ \phi(x_j^{(\alpha)} - \frac{i}{2}) \bar{\Psi}(x_1^{(\alpha)}, \dots, x_j^{(\alpha)} + i, \dots, x_L^{(\alpha)})$$

or its factorized form for each j, α :

$$t(v) Q(v) = \phi(v + \frac{i}{2}) Q(v - i) + \phi(v - \frac{i}{2}) Q(v + i)$$

$$\text{at } v = x_j^{(\alpha)}, \text{ any } j, \alpha$$

polynomiality $\Rightarrow$ valid for any $v \in \mathbb{C}$

We should be able to define the wave-function only at 2^L SoV points.

$$\text{if } x_j^{(\wedge)} = \theta_j \pm \frac{i}{2}, \text{ then } \phi(x_j^{(\wedge)} \mp \frac{i}{2}) = 0.$$

$$\begin{cases} t(\theta_j + \frac{i}{2}) \bar{\Psi}(\dots x_j = \theta_j + \frac{i}{2} \dots) = \phi(\theta_j + i) \bar{\Psi}(\dots x_j = \theta_j - \frac{i}{2} \dots) \\ t(\theta_j - \frac{i}{2}) \bar{\Psi}(\dots \theta_j - \frac{i}{2} \dots) = \phi(\theta_j - i) \bar{\Psi}(\dots x_j = \theta_j + \frac{i}{2} \dots) \end{cases}$$

$\Rightarrow$ determines $\bar{\Psi}$ at different x 's consistently

These SoV points are consistent w. construction of twisted B-op.

Q: Why $B(u)$ is nilpotent for periodic case?

$$[S^z, B] = -B, \quad \text{or} \quad B|J, j_z\rangle = |J, j_z-1\rangle$$

$$B^n|J, j_z\rangle = 0 \quad \text{for } n > 2J+1$$

Q: Why SoV basis useful?

Used for the resolution of identity $1 = \sum_x |x\rangle\langle x|$

Bethe states $|u\rangle \leadsto$ don't know all sol

Natural basis $|\uparrow\downarrow\rangle \leadsto$ generally complicated